

## The map — three diseases

- **Adverse selection:** hidden **type**, **before** the deal (pre-contract). The informed side decides whether to trade → the pool that shows up is biased against you.
- **Moral hazard:** hidden **action**, **after** the deal (post-contract). Once risk is shifted, behavior changes — drive worse, over-treat.
- **Signaling:** the informed side takes a **costly action** to prove its type. Works only if the action is **cheaper for good types**.
- **Timing test** (one-liner): does the problem bite **before** signing (who shows up?) or **after** (how do they behave once covered?). Real contracts (deductibles) often trigger both.

Common thread: private information makes competitive markets inefficient — trades that should happen don't, or the wrong ones do.

## Lemons market (Akerlof) — adverse selection

- Sellers know quality; buyers only know the **distribution** → buyers pay only for **average** quality.
- At an average price, **owners of the best cars refuse to sell** → pool worsens → price falls → more exit...



the unraveling loop — in the worst case only lemons trade

- **REE recipe:** (1) guess which types sell → (2) price =  $E[V]$  over that set → (3) check each type: sells iff price  $\geq$  its seller's own value → (4) iterate until the guess confirms itself; keep the **largest self-consistent set**.

all types trade  $\Leftrightarrow E[V | \text{all types}] \geq$  the best seller's reservation value

🔍 **Example — full unraveling.** 3 types, equal shares; (seller, buyer) values: Good (1000, 1500), Med (500, 750), Bad (100, 150).

- ① All trade:  $E[V] = \frac{1}{3}(1500+750+150) = 800 < 1000 \rightarrow$  Good exits.
- ② Med+Bad:  $E[V] = \frac{1}{2}(750+150) = 450 < 500 \rightarrow$  Med exits.
- ③ Bad only:  $150 \geq 100 \checkmark$ . **Price 150, only lemons trade.**

🔍 **Example — partial unraveling.** 5 Good, 20 Med, 5 Bad. All:  $E[V] = 775 < 1000 \rightarrow$  Good exits. Med+Bad pool is now 80% Med:  $E[V] = 0.8(750) + 0.2(150) = 630 \geq 500 \rightarrow$  Med & Bad both trade. Concentration of mid-quality saves the market.

Multiple equilibria can coexist ('only lemons' + 'all sell'). Fixes: certification, inspections, warranties ( $\rightarrow$  signaling).

## Contract vocabulary (used again in 2° pricing, p.2)

- **IR (participation):** the deal must beat the outside option, or that type won't sign.
- **IC (self-selection):** each type must prefer **its own** contract to mimicking another's.
- **Pooling vs separating:** one contract taken by all types vs a menu that splits them apart.

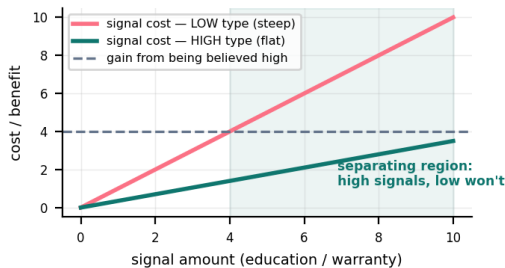
## Signaling &amp; screening (warranty · Spence)

- Talk is free — everyone **claims** to be good. A signal separates types only when **imitating it is too expensive for the low type** (single crossing).

separating equilibrium:  $c_H \leq P_H - P_L \leq c_L$

- **Left:** the high type wants to signal — the price/wage gain covers its (low) cost.
- **Right:** the low type refuses to mimic — the same gain doesn't cover its (high) cost.
- **Screening** = the **uninformed** side offers a menu so types self-select (IR + IC).

Single crossing: the signal is cheap only for the high type



🔍 **Example — warranty.** A warranty costs the good seller 300 but the lemon seller 1300. If being believed "good" is worth  $\sim 1000$ :  $300 \leq 1000 \leq 1300 \rightarrow$  only good cars carry one  $\rightarrow$  warranty = credible signal.

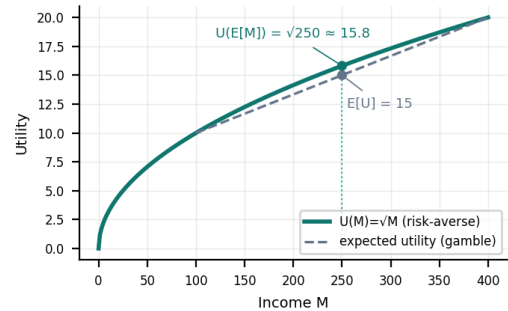
**Spence:** education with **zero** productivity gain still separates (easier for high-ability), wages track ability — but study costs are burned  $\rightarrow$  **signaling can be pure social waste**.

⚠ Always check **both** inequalities. If everyone signals — or nobody does — the equilibrium is **pooling** and the signal reveals nothing.

## Risk aversion &amp; insurance

- Concave  $U$  (e.g.  $U = \sqrt{M}$ )  $\Rightarrow E[U(M)] < U(E[M])$ : a risk-averse agent prefers the sure thing  $\rightarrow$  gladly pays an **actuarially fair premium** = expected covered loss ( $\pi = p \cdot L$ ; partial cover  $x$ :  $\pi = p \cdot x$ ). Will pay up to  $E[\text{loss}] +$  risk premium.

Concave  $U$ : sure thing beats the fair gamble  $\rightarrow$  buys insur



- **Observable types:** each group gets its own fair price  $\rightarrow$  everyone insures.
- **One pooled premium:** underpriced for the sick, overpriced for the healthy  $\rightarrow$  **the healthy walk away**  $\rightarrow$  pool worsens  $\rightarrow$  premium rises  $\rightarrow$  more leave: the **adverse-selection spiral**.
- **Market's fix — a menu:** full coverage @ high premium vs partial @ low. High-risks pick full, low-risks partial  $\rightarrow$  **separating equilibrium**; the healthy end up under-insured  $\rightarrow$  inefficient.

🔍 **Example — who stays in the pool?** Cost / reservation price: H 1000/2000 (25%), M 2000/3000 (50%), L 4000/6000 (25%).

- ① Uniform, all buy: avg cost =  $\frac{1}{4}(1000) + \frac{1}{2}(2000) + \frac{1}{4}(4000) = 2250 >$  H's RP 2000  $\rightarrow$  H drops out.
- ② M+L pool: avg =  $(\frac{1}{2} \cdot 2000 + \frac{1}{4} \cdot 4000) / 0.75 = 2667$ ; RPs 3000, 6000 both  $\geq 2667 \checkmark \rightarrow$  self-consistent. 25% uninsured.
- ③ **Subsidy** to bring H back:  $2250 - 2000 = 250$  per H (price falls to 2250 for M, L too).

## Moral hazard

- Insured people don't pay at the point of use  $\rightarrow$  consume past the point where value = cost.

uninsured:  $P = MC \rightarrow Q^* = D(MC)$  · insured:  $P = 0 \rightarrow Q^0 = D(0)$  ·  $DWL = \frac{1}{2} \cdot MC \cdot (Q^0 - Q^*)$

⚠ The fair premium must use **post-insurance usage**:  $\pi = E[MC \cdot Q^0]$ . Plugging in the old  $Q^*$  is the classic exam error.

- **Coinsurance (pay share  $\alpha$ ):** you face price  $\alpha \cdot MC \rightarrow$  usage lands **between**  $Q^*$  and  $Q^0$ . Less over-use, less risk protection.
- **Deductibles** curb moral hazard but mostly attract the healthy  $\rightarrow$  can make adverse selection **worse**. Full efficiency is out of reach.

🔍 **Example — expected spend, with vs without.**  $Pr(\text{mild, mod, sev}) = \frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ ; demands 5—P, 15—P, 20;  $MC = 2$ .

- ① Uninsured ( $P=2$ ):  $Q = 3, 13, 20 \rightarrow$  spend =  $2 \cdot [\frac{1}{2}(3) + \frac{1}{3}(13) + \frac{1}{6}(20)] = 18.33$ .
- ② Insured ( $P=0$ ):  $Q = 5, 15, 20 \rightarrow$  fair premium =  $2 \cdot [\frac{1}{2}(5) + \frac{1}{3}(15) + \frac{1}{6}(20)] = 21.67 > 18.33$ .

Gap = moral-hazard loss; units past  $Q^*$  are valued below  $MC \rightarrow DWL$ .

🔍 **Example — agency (physician vs HMO).** Treatment A: cost 1200,  $E[U]=2$ ; B: cost 2400,  $E[U]=15$ . Physician (ignores cost)  $\rightarrow$  B. HMO (utility per \$):  $15/2400 > 2/1200 \rightarrow$  also B here — but the **conflict of objectives** is the agency point.

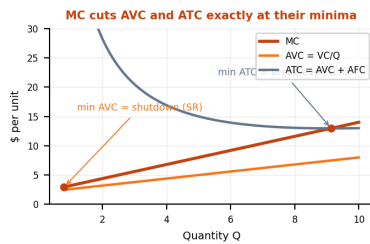
**Risk selection:** uniform regulated premiums  $\rightarrow$  funds compete by **benefit design that attracts the healthy**, not by price.

## Exam radar — asymmetric information

- **"Which types trade / who buys"**  $\rightarrow$  test pools for self-consistency (REE); the top type exits first.
- **"Uninsured"**  $\rightarrow P = MC$ . **"Fully insured"**  $\rightarrow P = 0$ , over-consume; fair premium on the **insured** quantities.
- **"Break-even premium"** = expected cost of those who **actually buy**, not the whole population.
- **Keyword  $\rightarrow$  model:** "can't tell quality" = lemons · "behavior after coverage" = moral hazard · "prove their type" = signaling.
- **Government subsidy**  $\rightarrow$  amount that makes the excluded type's  $RP \geq$  premium.

## Costs = opportunity costs

- **Economic profit** = revenue – all opportunity costs, implicit included (your time, forgone rent/wages) → smaller than accounting profit.
  - **Sunk costs are unrecoverable** → **ignore them** in every decision; only avoidable/marginal costs matter.
- $$TC = FC + VC \cdot MC = dTC/dQ \cdot AVC = VC/Q \cdot ATC = AVC + AFC \text{ (} AFC = FC/Q \text{)}$$



■ **Example — sunk & implicit costs decide.** Plane already flying (cost sunk); extra passenger MC = 0; offer 300 > 0 → **accept**, ignore the \$500 average cost. **Degree:** 35k tuition + 3k books + **40k forgone wages** = 78k/yr × 3 = 234k > benefit 5.5k × 40yr = 220k → don't study. **Derive:**  $TC = 5 + 40Q + 20Q^2$  →  $AVC = 40 + 20Q$ ,  $MC = 40 + 40Q$ ; ATC min at  $Q = 0.5$ , where  $MC = ATC$ .

## Why monopolies exist

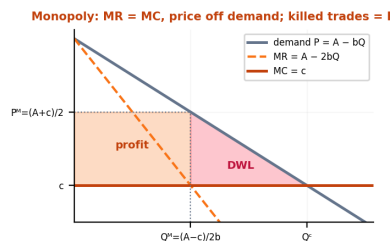
- **Barriers to entry:** a key resource owned by one firm · government-granted rights (patents, copyright, licences) · the production technology itself.
- **Natural monopoly:** huge fixed cost + small MC → ATC still falling at market size → one firm serves the market cheaper than two could (utilities, networks).

## Single-price monopoly — the pipeline

- **Why  $MR < P$ :** one more unit earns P but cheapens every unit already sold → the monopolist stops short of the efficient quantity.

$$P = A - bQ \Rightarrow MR = A - 2bQ \text{ (same intercept, twice the slope)}$$

$$MR = MC: Q^M = (A - c)/2b, P^M = (A + c)/2, \pi = (P^M - c) \cdot Q^M$$



■ **Example — rising MC.**  $P = 1300 - 5q$ ,  $MC = 50 + 10q$ . Set  $MR = MC$ :  $1300 - 10q = 50 + 10q \rightarrow 20q = 1250 \rightarrow q^* = 62.5$ ; read price off demand:  $P^* = 1300 - 5(62.5) = 987.5$ .

## Revenue maximisation &amp; capacity

- Max **revenue** (not profit) where  **$MR = 0$**  ( $|E| = 1$ ). If demand at that price exceeds capacity, sell out at capacity (corner).
- **Example — stadium.**  $P = 40 - 0.5Q$  (Q in k seats), capacity 56k, current price \$10 → demand 60k → sell 56k,  $TR = 560k$ . Rev-max:  $MR = 40 - Q = 0 \rightarrow Q = 40k$ ,  $P = \$20$ ,  $TR = 800k$  → ticket was **underpriced**; optimum leaves 16k seats empty — the cost of one uniform price.

## Elasticity, Lerner index &amp; taxes

$$MR = P \cdot (1 - 1/|E|) \cdot L = (P - MC)/P = 1/|E| \cdot \text{constant-} |E|: P^* = MC / (1 - 1/|E|)$$

- **Markup mirrors elasticity:** captive (inelastic) buyers → high markup; buyers with substitutes (elastic) → low. Competitive firm:  $L = 0$ .  $|E|$  is bigger with close substitutes & in the long run.
- **Tax logic (the OPEC question):** a **profit tax** (share of  $\pi$ ) leaves (P,Q) unchanged — maximizing 10% of profit = maximizing profit. A **per-unit tax t** acts like  $MC \uparrow \rightarrow P \uparrow, Q \downarrow, DWL \uparrow$ . Incidence (demand slope b, MC slope m):  $\Delta P/\Delta t = b/(2b + m)$ .
- **Example — pricing off  $|E|$ .**  $MC = 16$ ,  $|E| = 5 \rightarrow P^* = 16/(1 - 1/5) = 20$ . Students  $|E| = 9 \rightarrow P^* = 18$  (the "student discount" is elasticity, not kindness).  $|E| = 2 \rightarrow L = 1/2 \Rightarrow P = 2 \cdot MC$ .
- **Example — per-unit vs lump-sum.**  $P = 1300 - 5q$ ,  $MC = 50 + 10q$  ( $b=5$ ,  $m=10$ ).  $t = 300$ :  $q: 62.5 \rightarrow 47.5$ ,  $P: 987.5 \rightarrow 1062.5$ ; pass-through  $5/20 = 25\%$ . Lump-sum 5000:  $q, P$  unchanged. **T/F:**  $P = 100 - 10Q \rightarrow MR < 0$  past  $Q = 5 \rightarrow$  never sells > 5m units, **regardless of costs**.

## Bundling (multi-product monopoly)

- Bundle when customers **disagree about which good is worth more** — one package price captures surplus that separate prices leak.
- **bundling profitable** ⇔ **reservation prices negatively correlated**
- **Method:** profit under (i) best **separate** prices, (ii) best **pure bundle** price, (iii) **mixed** bundling / versioning (bundle + stand-alones) → walk candidate prices,  $\pi = (\# \text{buyers}) \cdot (\text{price} - MC)$ , pick the max.
- **Example — Excel/Word.** WTPs (90,10), (80,40), (40,80), (10,90). Separate best ≈ **320**; pure bundle @100 →  $4 \cdot 100 = 400$ ; mixed (bundle @120 + single @90) → **420**. **Season tickets** (0 cost): separate = 44; pure bundle @23 = 46; versioned (full @26 + 2-concert pass @23) = **49** — mixed/versioning wins again.

△ Same-direction rankings (positive correlation) → bundling adds nothing; check the table first.

## Price discrimination — capturing surplus

- **Needs:** market power · segments with different  $|E|$  · a way to infer WTP · **no arbitrage**. Blockers: warranty void on resale · personal services · coupons/rebates · hardcover-then-paperback · regional versions.
- **1° — perfect / two-part tariff:** per-unit  $p = MC$  (quantity efficient, zero DWL) + fixed fee  $T = \text{entire CS}$ ; buyer breaks even. Separate fee per buyer when types are known.
- **free-units card** ( $p = 0$ ): ① write  $p(Q)$ ,  $MC(Q)$  ②  $p(Q) = MC(Q) \rightarrow Q^* \Rightarrow T = [p(0) + p(Q^*)]/2 \cdot Q^*$  work in  $Q - p(Q)$ ,  $MC(Q)$  eat  $Q$ , never flip to P

■ **Example — discriminating TPT.**  $C1: q = 40 - 2P$ ,  $C2: q = 20 - P$ ,  $MC = 6$ . Set  $p = 6$ :  $q_1 = 28$ ,  $q_2 = 14$ . Fees = each CS (intercepts 20):  $T_1 = 1/2(20 - 6)(28) = 196$ ,  $T_2 = 1/2(20 - 6)(14) = 98 \rightarrow \pi = 294$  (all surplus taken).

- **2° — hidden types → menu design:** buyers self-select. Low type: **IR binds** (full WTP, zero surplus). High type: **IC binds** — indifferent to mimicking → keeps an **information rent**. "No distortion at the top": high type gets  $p = MC$ ; **shrink the low type's bundle** while rent saved > surplus lost; stop when equal ( $a = b$ ). Extreme: serve only the high type.

$$\text{balance the gaps: shrink } q_L \text{ until } AR_{low} - MC = AR_{high} - AR_{low}$$

■ **Example — menu (Ex 5 Q3e).**  $AR_{high} = 20 - q/2$ ,  $AR_{low} = 20 - q$ ,  $MC = 6$ . High: efficient  $q_H = 28$ . Low:  $14 - q_L = q_L/2 \rightarrow q_L = 9.33$ ,  $p_L = 10.67$ . Rent =  $1/2(9.33)(4.67) = 21.78 \rightarrow \pi = 196 + 87.11 - 21.78 = 261.33$  (between uniform and 1°).

- **3° — observable groups** (students, regions, peak/off-peak): set  $MR_i = MC$  in each segment; resale blocked.  $P_i = MC/(1 - 1/|E_i|) \rightarrow$  **the inelastic group pays more**. Shared rising MC → MC depends on total output.

■ **Example — split the market.**  $Q = 500 - P$ ,  $MC = 100$ . Uniform:  $Q = 200$ ,  $P = 300$ . Split: business 110 @375, students 90 @250 (Q unchanged, CS falls). **Capped/gov buyer + rising MC:**  $P = 100 - X$ ,  $MC = X$ ; fill gov cap 10 @50, then  $100 - 2X_p = X_p + 10 \rightarrow X_p = 30$ ,  $P_{pub} = 70$ .

- **Many markets, one price — kinks & corners:** no discrimination → **sum demands horizontally**. run  $MR = MC$  on the sum, then **check every group buys** ( $p < \text{each choke}$ );  $q_i < 0$  → corner: drop that market, re-solve, compare with serving the strong market alone. n identical convex plants:  $TC(Q) = n \cdot TC_1(Q/n)$ ; avoidable  $F \Rightarrow$  serve a market only if its  $\pi \geq 0$ .

## Uniform tariff &amp; the value of information

- Can't tell types apart → **one** two-part tariff for all: fee pinned by the **smaller** buyer's surplus,  $T = CS_{small}(p)$ ; raise p above MC to claw margin from the big buyer; maximize  $\pi$  over p. (Or serve only the high type with  $T = CS_{high}$  — compute both, keep the max.)
- **value of type info =  $\pi(\text{discriminate}) - \pi(\text{one tariff})$  — a difference, not a level**
- **Example.**  $C1: q = 40 - 2P$ ,  $C2: q = 20 - P$ ,  $MC = 6$ . Uniform TPT:  $T = 1/2(20 - p)^2$ ,  $\pi(p) = 2T + (p - 6)(60 - 3p) \rightarrow p^* = 9.5$ ,  $T = 55.1 \rightarrow \pi = 220.5$  vs 294 discriminating → info worth **73.5**. Also:  $q_1 = 9 - p$ ,  $q_2 = 12 - p$ ,  $MC = 2 \rightarrow$  discriminate 74.5, uniform  $p^* = 3.5 \rightarrow 51.25$ ; info worth **23.25**.

## Welfare bookkeeping (used everywhere)

- **CS** = under demand, above price ( $= 1/2(P_{max} - p)Q$  linear) · **PS** = above MC, below price · **DWL** = value-above-cost trades that don't happen. Any wedge  **$P > MC$**  creates DWL — monopoly, taxes, double markups; "compare regimes" = who holds the surplus, how much leaked?
- **Welfare of PD:** 1° is efficient (surplus transferred, not destroyed); 3° and menus raise profit but move welfare **either way** — never assume "discrimination = lower total surplus".

## Exam radar — pricing

- **"Sold separately / single price"** → walk the WTP ladder;  $\pi = n \cdot (p - MC)$ .
- **"Membership card / two-part tariff"** →  $p = MC$ , fee = CS; never price at 0 if MC rises.
- **"Value of identifying types"** →  $\pi_{disc} - \pi_{uniform}$ .
- **"Serve one or both?"** → compare with serving only the big buyer; drop the weak one when it drags the shared fee down.
- **Per-unit vs lump-sum tax** → per-unit shifts MC (pass-through  $b/(2b + m)$ ); lump-sum/profit tax leaves P, Q unchanged. **"Accept / continue?"** → avoidable costs only.

**Bertrand — price war, identical good**

- Any price above cost invites undercutting by a hair → only rest point:  **$P = MC$ , zero profit, with just two firms** — the Bertrand paradox. Escapes: differentiation, capacity limits, repetition.
- Unequal costs ( $c_1 < c_2$ ):** the cheap firm serves the whole market. Undercut only if you must — if its monopoly price already sits below  $c_2$ , it just charges that.

$$p^* = \min(p^m(c_1), c_2 - \epsilon) \cdot \text{entry: enter iff post-entry } \pi \geq F$$

- Entry cost F:** symmetric Bertrand → post-entry  $\pi = 0$  → **nobody enters**; the incumbent prices as a monopolist.

Example — which regime?  $P = 100 - Q$ ,  $MC_1 = 10$ ,  $MC_2 = 80$ . Firm 1's monopoly price =  $55 < 80$  → charge **55**,  $\pi = 2025$ , firm 2 out. If  $MC_2 = 50 < 55$  → limit-price at **50**,  $\pi = 2000$ . **Cutoff form:**  $Q = 10 - P$ ,  $MC_1 = 6$ ,  $MC_2 = c$ : if  $2 < c < 6$ , firm 2 limit-prices at 6,  $\pi_2 = 4(6 - c)$ ; if  $c < 2$ , its monopoly price  $(10 + c)/2 < 6$  → just monopolise.

**Differentiated products — softer price war**

$$q_1 = A - 2p_1 + p_2 \text{ (substitutes: the rival's high price feeds you)}$$

- Pipeline:**  $\pi_1 = (p_1 - k_1)q_1$  → FOC  $\partial \pi_1 / \partial p_1 = 0$  → best response  $p_1(p_2)$  → same for firm 2 → solve together (use symmetry). BRs slope **up**: prices are **strategic complements**. Differentiation blunts undercutting → positive profits survive.
- Merger** internalises the cross-price effect → raises both prices & profits.

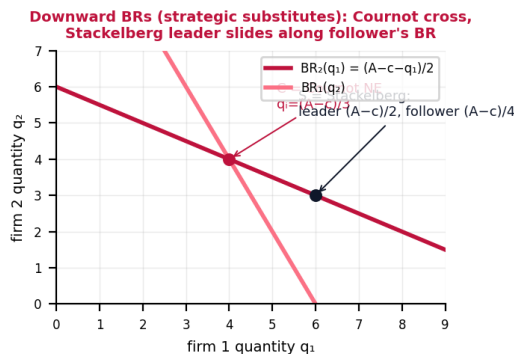
Example.  $q_1 = 168 - 2p_1 + p_2$ ,  $c = 0$ . FOC:  $P_1 = (168 + P_2)/4$ ; symmetry  $P = 56$ ,  $q = 112$ ,  $\pi = 6272$ . Merger:  $P = 84$ ,  $\pi/\text{firm} = 7056$  — both gain.

Example — choosing the mode.  $p_1 = 24 - q_1 - 0.5q_2$ ,  $MC = 0$ . Cournot  $\pi = 92.16$ ; Bertrand  $\pi = 85.33$ . In the  $2 \times 2$  mode game **Quantity is dominant** → NE = (Q, Q) = Cournot (softer competition).

**Cournot — competing in quantities**

- Each firm commits to output; price clears at  $P = A - Q$ . Extra units depress the **shared** price → self-restraint, but less than a cartel's.

BRs:  $q_1 = (A - c - q_2)/2 \Rightarrow q_1 = (A - c)/3$ ,  $P = (A + 2c)/3$ ,  $\pi_1 = (A - c)^2/9$   
**unequal costs:**  $q_1 = (A - 2c_1 + c_2)/3$ ,  $P = (A + c_1 + c_2)/3$ ,  $\pi_1 = q_1^2 \cdot n$  firms:  $q_1 = (A - c)/(n + 1)$ ,  $P = (A + nc)/(n + 1) \rightarrow MC$



Example — convex costs.  $P = 550 - Q$ ,  $MC_1 = q_1$ ,  $MC_2 = 2q_2$ . FOCs:  $550 - 3q_1 - q_2 = 0$  and  $550 - q_1 - 4q_2 = 0 \Rightarrow q_1 = 150$ ,  $q_2 = 100$  (cheaper firm takes the larger share). Symmetric check:  $P = 120 - Q$ ,  $MC = 70 \Rightarrow q_1 = 50/3 \approx 16.7$ ,  $P \approx 86.7$ ,  $\pi \approx 277.8$ .

△ Rising MC ( $TC = q^2$ ,  $0.5q^2 \dots$ )? The boxed results assume  $MC = c$ . **Rebuild:**  $\pi = (A - Q)q_1 - TC(q_1) \rightarrow$  FOC  $\rightarrow$  BRs (slopes now differ)  $\rightarrow$  solve.

**Stackelberg — the value of moving first**

- Solve backwards:** substitute the follower's BR into the leader's profit, then maximize. Commitment is the trick: the leader floods the market, the follower shrinks; total Q rises vs Cournot → consumers gain.

$$q_1 = (A - c)/2, q_2 = (A - c)/4, P = (A + 3c)/4 \cdot \pi_1 = (A - c)^2/8 > \text{Cournot } (A - c)^2/9 > \pi_2 = (A - c)^2/16$$

- Non-constant MC** ( $MC_i = c_i + dq_i$ ) — verification shortcut with  $s = 2 + d_2$ :  $q_1 = [s(A - c_1) - (A - c_2)] / [s(2 + d_1) - 2]$ ,  $q_2 = (A - c_2 - q_1)/s$ . General route: follower FOC  $\rightarrow q_2(q_1)$  with slope  $dq_2/dq_1$ ; leader FOC:  $A - 2q_1 - q_2 - q_1(dq_2/dq_1) = MC_1$ .

Example (Ex 8 Q3b).  $P = 550 - Q$ ,  $MC_1 = q_1$ ,  $MC_2 = 2q_2 \Rightarrow s = 4$ :  $q_1 = (4 \cdot 550 - 550)/(4 \cdot 3 - 2) = 1650/10 = 165$  (vs Cournot 150),  $q_2 = 385/4 = 96.25$  (vs 100) — convex costs mute the first-mover shift.

△ Leader advantage needs an **irreversible commitment** — else it's just Cournot.

**Cartel, cheating on it & buying the rival**

- A cartel maximises **joint** profit (acts as one monopolist:  $Q = (A - c)/2$ , split) — jointly best but **not a NE**: best-responding to the rival's cartel share pays more.

**cheat**  $q = 3(A - c)/8$ ,  $\pi = 9(A - c)^2/64$ : **cheat > collude**  $(A - c)^2/8 > \text{Cournot } (A - c)^2/9 \rightarrow$  cartels self-destruct  
**buy the rival (2→1):** max pay =  $\pi^m - \pi^c = 5(A - c)^2/36 > \text{min ask} = \pi^c = 4(A - c)^2/36$

Example.  $P = 200 - Q$ ,  $MC = 80$ . Cournot:  $q_i = 40$ ,  $\pi = 1600$  each. Cartel:  $q = 30$ ,  $\pi = 1800$ . Cheat on it: BR to 30  $\rightarrow q = 45$ ,  $\pi = 2025$  (cheat > cartel > Cournot). Monopoly  $\pi = 3600 \rightarrow$  buyer pays  $\leq 3600 - 1600 = 2000$ ; seller accepts  $\geq 1600$ .

A 2→1 merger always closes (ceiling > floor). But merging 2 of 3 firms can LOWER joint profit — the outsider free-rides. Compute, don't assume. Escapes from the dilemma: repeated play (tit-for-tat / folk theorem), binding contracts.

**One market, four regimes ( $P = A - Q$ ,  $MC = c$ )**

| Regime                 | Q            | P            | industry $\pi$  |
|------------------------|--------------|--------------|-----------------|
| Cartel / monopoly      | $(A - c)/2$  | $(A + c)/2$  | $(A - c)^2/4$   |
| Cournot (2)            | $2(A - c)/3$ | $(A + 2c)/3$ | $2(A - c)^2/9$  |
| Stackelberg            | $3(A - c)/4$ | $(A + 3c)/4$ | $3(A - c)^2/16$ |
| Bertrand / perf. comp. | $A - c$      | $c$          | 0               |

Going down:  $Q \uparrow$ ,  $P \downarrow$ , industry  $\pi \downarrow$ , welfare  $\uparrow$  — this ordering answers half the comparison MCs.

**Perfect competition — the benchmark (SR & LR)**

- Price takers:** no firm-level demand curve, **no MR**. Each supplies where  **$P = MC$** , above min AVC (below  $\rightarrow$  shut down);  $D = S$  maximizes  $CS + PS$ .

① flip  $P = MC$  into  $q_i(P)$  ② add quantities:  $Q^s = n \cdot q_i(P)$  ③ clear  $Q^s = Q^d(P) \rightarrow P$   
 step ③ only P allowed · multiply quantities by n, not MC · profit uses FULL TC

- Long run:** profit attracts entry  $\rightarrow$  supply shifts out  $\rightarrow \pi = 0$ ,  **$P = \min ATC$** , each firm at efficient scale; demand sets only the **number of firms**, not the LR price. A demand fall  $\rightarrow$  same P, fewer firms; a cost cut  $\rightarrow$  lower LR price.

- Two activities coexist**  $\rightarrow$  equate **net** returns; gross-return gap = cost difference (compensating differential).

Example. Demand  $P = 1000/X$ ,  $C = 12.5 + 0.5x^2$  ( $MC = x$ ), 10 firms. SR:  $x = P \rightarrow X = 10P$ ; clear:  $P = 1000/10P \Rightarrow P^2 = 100 \Rightarrow P = 10$ ,  $x = 10$ ,  $\pi = 100 - 50 - 12.5 = 37.5$ . LR: min AC where  $MC = AC \Rightarrow x = 5$ ,  $AC = 5 \Rightarrow P = 5$ ,  $X = 200 \rightarrow 40$  firms.

Example — Airbnb vs long-let. Furnished short-let needs 10k/yr furniture. Coexistence  $\Rightarrow R_{\text{Airbnb}} - 10k = R_{\text{long}} \rightarrow$  gross gap = **10k**.

**Vertical chains — double marginalisation**

(Producer ( $MC = k$ ))  $\rightarrow w > k$  (Retailer)  $\rightarrow p > w$  (Consumers)

- Solve backwards:** retailer max  $(p - w)Q \rightarrow MR_r = A - 2Q = w$ . That equation **is** the producer's demand curve; producer sets  $MR_p = A - 4Q = k$ .

$$Q = (A - k)/4, w = (A + k)/2, p = (3A + k)/4 \cdot \pi_{\text{prod}} = (A - k)^2/8, \pi_{\text{ret}} = (A - k)^2/16 < \text{integrated } (A - k)^2/4$$

- Two stacked markups  $\rightarrow$  output **half** the integrated monopolist's; chain profit AND CS shrink — everyone loses. **Fixes:** vertical integration · **two-part tariff** ( $w = k +$  franchise fee  $\rightarrow$  retailer picks monopoly Q, fee splits the pie) · RPM.

Example.  $P = 200 - Q$ ,  $k = 0$ . Integrated:  $Q = 100$ ,  $P = 100$ ,  $\pi = 10000$ . Chain:  $w = 100$ ,  $Q = 50$ ,  $p = 150 \rightarrow \pi = 5000 + 2500 = 7500 < 10000$ . TPT ( $w = 0 +$  fee) restores 10000.

**Exam radar — competition**

- Asymmetric Bertrand**  $\rightarrow$  efficient firm prices at min(rival's MC, own monopoly price).
- "Sells through a distributor"**  $\rightarrow$  backward induction;  $\pi_{\text{chain}} < \pi_{\text{integrated}}$ ; fix with TPT/merger.
- "Chooses first"**  $\rightarrow$  Stackelberg (substitute follower's BR). **"Simultaneous quantities"**  $\rightarrow$  Cournot.
- "Max price to acquire a rival"** = monopoly  $\pi$  — buyer's duopoly  $\pi$ ; min = seller's duopoly  $\pi$ .
- "Long-run, free entry"**  $\rightarrow P = \min AC$ ,  $\pi = 0$ ; demand sets only the number of firms.
- Differentiated mode choice**  $\rightarrow$  Quantity dominates Price  $\rightarrow$  firms land at Cournot.



## What is a game?

- ▲ A **game** = players + strategies + payoffs. Reach for it whenever **my best move depends on what you do** — firms in one industry, bidders, cartel partners.
- **Normal form** = payoff matrix; the **row player's payoff is written first**. Read it: freeze the rival's choice (one column) → scan your rows for your best payoff → swap roles.
- **Simultaneous** moves → solve with NE. **Sequential** (Stackelberg, entry, producer→retailer): later movers see earlier moves → **solve backwards**: last mover's reaction folded into the first mover's problem.

## Dominance — the first filter

- A **strictly dominates B** if A pays more against **every** rival choice; a rational player never plays a strictly dominated strategy. **Weak**:  $\geq$  everywhere,  $>$  somewhere (say which you use).
- **IESDS**: cross out dominated strategies → game shrinks → new ones may become dominated → repeat. **Safe facts**: with strict dominance the deletion order never matters, and **no NE is ever eliminated** — if one cell survives, it is the unique NE.

▲ A dominant-strategy equilibrium is automatically a NE — but most NE involve no dominance at all. Don't equate the two.

## Nash equilibrium — the workhorse

- **NE**: no player gains by deviating **alone**, holding others fixed = everyone plays a **best response**: each  $s_i \in BR_i(s_{-i})$ . Self-enforcing — no one regrets their choice. Every market model on p.2-3 is "find the NE" in disguise.
- **Find all pure NE (dot method)**: dot the row player's best payoff in **each column**, the column player's best in **each row**; a cell with **both dots** = pure NE.

|            | Don't hire | Hire    |
|------------|------------|---------|
| Don't hire | 5, 5       | 3, 6 •  |
| Hire       | •6, 3      | •4, 4 • |

Unique NE: (Hire, Hire) → (4,4) even though (5,5) was on the table. No one deviates alone — that is the prisoner's dilemma.

🔍 **Example — dot method.** U: (3,2)(1,4) / D: (2,1)(4,3). Col L → U best (3>2); col R → D (4>1). Row U → R (4>2); row D → R (3>1). Both dots land on (D,R) = (4,3).

## Parametric NE conditions

- **"For which values of a, b... is X a NE?"** → impose the two no-deviation (BR) inequalities on the candidate cell. For **uniqueness**, also negate every other profile's conditions.
- 🔍 **Example.** Rows U/D, cols L/R; U: (c,1)(1,a), D: (d,1)(2,b). (D,R) is the **unique** NE iff **b < 1** (kills (D,L)), **d ≥ c** (D is a BR), and **(d > c or a > 1)** (kills (U,R) profiles).

## Unravelling &amp; the commons trick

- **Traveller's dilemma**: reports 180-300; lower report +5, higher -5. (300,300) is Pareto-best but undercutting by  $\epsilon$  always pays → unravels to unique NE (180,180). Rational  $\neq$  efficient — same lesson as the PD. Escapes: reputation, contracts, social preferences.
- **Commons game (Ex 8 Q5), payoffs off the FOC**:  $TC_i = (q_i + Bq_j)^2$ ,  $U_i = q_i - TC_i$ . **Don't expand the square** — each player's own FOC fixes its bracket: follower  $q_2 + Bq_1 = \frac{1}{2} \rightarrow TC_2 = \frac{1}{4}$ ; leader  $q_1 + Bq_2 = \frac{1}{2[1-B^2]} \rightarrow TC_1 = \frac{1}{4[1-B^2]^2}$ .
- 🔍 **Example (B = 1/2)**.  $q_1 = 5/9$ ,  $q_2 = 2/9$ .  $U_1 = 5/9 - 1/[4(3/4)^2] = 5/9 - 4/9 = 1/9$ .  $U_2 = 2/9 - 1/4 = -1/36$  — the follower can go negative; the leader's bracket is bigger, yet it still wins.

## Exam radar — games &amp; complements

- **"Odd/even, matching pennies"** → no pure NE, unique **mixed** NE ("no NE" is the trap). **BoS** → 3 equilibria; **RPS** → uniform mixed  $\frac{1}{3}$  each.
- **Test MC claims in order**: dominance first (strongest claim, dies fastest) → count pure NE → then mixed. "Unique NE" can mean the mixed one — when an MC says "NE" it **always includes mixed**.
- **"For which values... is it a NE"** → BR inequalities; for uniqueness negate the others.
- **Mixed NE** → make the **opponent** indifferent; solve for your own probability.
- **"Toll booths / console+game / supply chain"** → complements: more firms → higher total price.
- **Open-question pipeline**: ① demand & costs → ② each player's profit → ③ FOC (MR = MC / BR) → ④ solve for Q, P → ⑤  $\pi$ , CS, DWL → ⑥ compare regimes: who gains, who loses, where surplus leaked. Sequential → backward induction. Same recipe for non-market payoffs (commons).

## The classic 2x2 games — know the counts

## Battle of Sexes

|     |     |
|-----|-----|
| 2,1 | 0,0 |
| 0,0 | 1,2 |

2 pure NE + 1 mixed

## Coordination

|     |     |
|-----|-----|
| 3,3 | 0,0 |
| 0,0 | 1,1 |

(A,A) better; (B,B) safer

## Matching pennies

|      |      |
|------|------|
| 1,-1 | -1,1 |
| -1,1 | 1,-1 |

no pure NE → mix  $\frac{1}{2}/\frac{1}{2}$

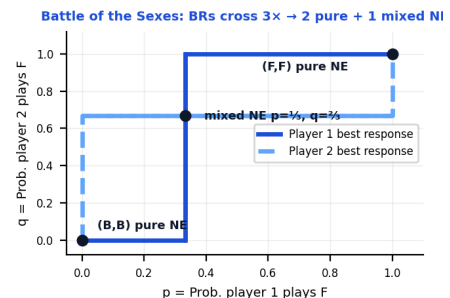
- **Prisoner's dilemma** (marketing/cartel game): dominant strategies → jointly bad outcome. 1 NE.
- **Battle of Sexes**: coordinate but disagree where → 2 pure + 1 mixed (sometimes miscoordinate). **Coordination/network**: payoff-dominant vs risk-dominant/safe — both NE; lock-in. Adoption bonus c: (A,A) is a NE iff  $c \geq b$ ; if  $a = b < c$  both (A,A) & (B,B).
- **Zero-sum** (pennies, RPS): my win = your loss → nothing to coordinate on → no pure NE; RPS mixes  $\frac{1}{3}$  each.

## Mixed strategies

- Used when **no pure NE exists** — and to find the **extra** equilibrium in coordination games.
- **The logic**: you only randomize if you're **indifferent** → in equilibrium each player's mix makes the **RIVAL indifferent** between her strategies. One indifference equation per player; solve the pair.

**choose your p so that:  $E[\text{rival's payoff} \mid \text{her } s_1] = E[\text{rival's payoff} \mid \text{her } s_2]$**

▲ Your mixing probability comes from **the rival's payoffs, not your own**. The single most-missed MC point in game theory.



🔍 **Example.** U: (0,2)(3,0) / D: (2,0)(0,3); let  $q = \Pr(P2 \text{ plays L})$ . P1's EU(U) =  $0 \cdot q + 3(1-q)$ ; EU(D) =  $2q$ . Indifference:  $3-3q = 2q \rightarrow q = \frac{3}{5}$ ; same for P2:  $2p = 3(1-p) \rightarrow p = \frac{3}{5}$ . In BoS (2,1): each puts  $\frac{3}{5}$  **on their own favorite** — and the mixed NE pays less than either pure NE, because they sometimes miss.

Nash's theorem: every finite game has at least one equilibrium, possibly mixed — so "this game has no NE" is never the right answer.

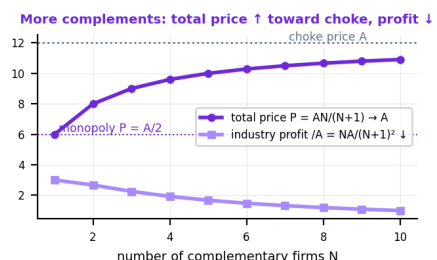
## TOPIC 4 — COMPLEMENTARY GOODS

one system, many sellers

## Perfect complements — N-firm pricing

- Buyer needs **all** components (console+game, toll booths on one road) → only the **total price**  $P = \sum p_i$  matters; demand  $Q = A - P$  (one shared Q, 1:1 sets).
- Each markup taxes **everyone's** demand — a negative pricing externality nobody internalizes: **double marginalisation, sideways** (same disease as the vertical chain, p.3).

**FOC firm i:  $A - P - p_i = 0 \Rightarrow p_i = A/(N+1)$ ,  $P = AN/(N+1)$ ,  $Q = A/(N+1)$ ,  $\pi_i = A^2/(N+1)^2$**   
**monopoly (N=1):  $P = A/2$  as  $N \rightarrow \infty$ :  $P \rightarrow A$  (choke),  $Q \rightarrow 0$**



🔍 **Example — toll road.**  $A = 10$ ,  $N = 2$ :  $p_i = 10/3$ ,  $P = 20/3 \approx 6.67$  > monopoly's 5;  $Q = 10/3$ ,  $\pi_i = 100/9 \rightarrow$  joint  $200/9 <$  monopoly 25. **Adding a firm hurts**:  $A = 12$ :  $N=1 \rightarrow P=6$ ,  $\pi=36$ ;  $N=3 \rightarrow P=9$ ,  $Q=3$ ,  $\pi_{tot}=27 < 36$ .

## Fixes &amp; the sign flip

- **Internalise the externality**: vertical/horizontal integration (→  $P = A/2$ ), two-part tariffs, revenue sharing — coordinate the **total** price. More complementary firms → higher P, lower Q, lower  $\pi$  — the exact opposite of substitutes.
- **Read the cross-price sign first**: substitutes compete → more firms lower P, welfare ↑. Complements → more firms raise P, welfare ↓.

▲ Substitutes: a merger **raises** prices. Complements: a merger **lowers** the total price and helps consumers. This sign flip is a favorite MC question.